

Markscheme

Mathematics: analysis and approaches

Practice question bank

30. (a)

Consider the function $f(x) = \frac{3x^2 - 2x + 5}{(x - 1)(x^2 + x + 1)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x - 1} + \frac{Bx + C}{x^2 + x + 1}$.

attempt to write $A(x^2 + x + 1) + (Bx + C)(x - 1)$ (M1)

$$3x^2 - 2x + 5 = A(x^2 + x + 1) + (Bx + C)(x - 1) \quad \text{A1}$$

$$\text{substituting } x = 1: 6 = 3A \Rightarrow A = 2 \quad \text{A1}$$

comparing coefficients of x^2 and constant terms (M1)

$$3 = A + B \Rightarrow B = 1 \quad \text{A1}$$

$$5 = A - C \Rightarrow C = -3 \quad \text{A1}$$

[6 marks]

(b)

Hence, by differentiating your partial fraction form term by term, find $f'(2)$.

differentiate each term (quotient rule on the second term) (M1)

$$f'(x) = -\frac{2}{(x - 1)^2} + \frac{-x^2 + 6x + 4}{(x^2 + x + 1)^2} \quad \text{A1}$$

$$f'(2) = -2 + \frac{12}{49} = -\frac{86}{49} \approx -1.755 \quad \text{A1}$$

[3 marks]